Shreya Jaiswal

Bachelor of Technology
Electrical Engineering
Indian Institute Of Technology, Ropar

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Portfolio | GitHub
LinkedIn
LeetCode

Education

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.43 (Till 6th Sem)	2023-2027
Senior Secondary	Central Board of Secondary Education	94.6%	2022
Secondary	Central Board of Secondary Education	85.2%	2020

Experience

- **Chitwan** *May 2026 – July 2026*
  Software Developer Intern Remote
  – Implemented room creation, shareable-link joining and turn validation for the multiplayer layer of a browser-based games platform (**6 games, up to 6 players per room**).
  – Integrated REST APIs and wrote client-side JavaScript for player, room, match and leaderboard data, with input validation and error handling.
- **Texas Instruments** *May 2025 – June 2025*
  WiSH 2025 Mentee On-site
  – Designed and tested analog and embedded-system prototypes using **CCS, QUCS, MSPM0, ASLK Kit, and ESP32**, including circuit simulation, sensor interfacing, and firmware development.

Projects

- **Velocity Keys** *May 2026 – Aug 2026*
  Full Stack Web Development **Website** | **GitHub**
  – Designed the real-time layer with **Socket.IO**: race rooms, per-tick progress sync, finish ordering, friend presence and direct challenges, with server-side state as the source of truth and recovery on mid-race disconnects.
  – Built the backend in **Node.js**/**Express** with **JWT** + **bcrypt** auth and a **PostgreSQL** schema via Prisma; front end in **Next.js, React, TypeScript, Tailwind**. Deployed on Render.
- **PCB Optical Inspection System** *July 2026 – Sep 2026*
  Full Stack,Machine Learning **Live** | **GitHub**
  – Built a **FastAPI** + **SQLAlchemy** inspection service (`/api/inspect`, history, PDF report endpoints) and a **React**/**TypeScript** dashboard with upload, annotated results, history and defect analytics.
  – Containerized with **Docker**, CI on **GitHub Actions**, deployed to **Google Cloud Run**; detection by a **YOLO11** model (95.8% precision / 94.0% recall on DeepPCB) with ORB + RANSAC registration.
- **FraudGuard – Fraud Scoring API** *May 2026 – Aug 2026*
  Machine Learning **GitHub**
  – Built a **FastAPI** scoring service (single and batch CSV endpoints, Pydantic validation, prediction history) and a **Streamlit** dashboard; **Docker Compose**, **55 pytest tests**, GitHub Actions CI with lint and smoke tests.
  – Trained an **XGBoost** model on 6.36M PaySim transactions with a leakage-safe chronological split and TreeSHAP reason codes on every prediction.
- **PMU Desktop Application – IEEE C37.118 Simulator** *Jan 2026 – May 2026*
  Systems, Networking **GitHub**
  – Implemented the **IEEE C37.118** binary protocol (config / header / data frames) over **TCP** in **Node.js**, streaming 8 channels at 60 fps to a PDC with an IP allow-list and injectable fault events; packaged as an **Electron** desktop app.

Technical Skills

- **Languages & Frameworks**: C, C++, Python, JavaScript, TypeScript, React.js, Next.js, Node.js, Express.js, FastAPI
- **Databases & Tools**: PostgreSQL, MongoDB, Firebase, Prisma, SQLAlchemy, Socket.IO, Tailwind CSS, Git, Docker
- **ML & Computer Vision**: PyTorch, YOLO, OpenCV, XGBoost, scikit-learn, SHAP, NumPy, Pandas, Streamlit

Key courses taken

- **CSE And Maths**: Introduction to Data Structures, Linear Algebra, Differential Equation, Calculus, Probability
- **Electrical & Electronics**: Digital Circuits, Signals and Systems, Control Engineering, Electromechanics, Communication Engineering, Electromagnetics, Power Electronics

Positions of Responsibility

- **Coordinator**, Aeromodelling Club, IIT Ropar *Apr 2024 - May 2025*
- **Co-Head**, Event Management, Advitiya'25, IIT Ropar *Dec 2024 - March 2025*

Miscellaneous

- **LeetCode**, Solved **400+ problems** focused on data structures, algorithms, and coding efficiency improvement
- **Codeforces**, Max Rating of 1160 | Handle: **Shreya246**